

REPORTING MISCONDUCT: THE ROLE OF PENALTY SEVERITY AND PERCEIVED FAIRNESS

Quy Lam
Ketki Sheth

August 2026

WORKING PAPER #2026-02

WORKING PAPER SERIES
DEPARTMENT OF ECONOMICS
HASLAM COLLEGE OF BUSINESS
<http://econ.bus.utk.edu>



Reporting Misconduct: The Role of Penalty Severity and Perceived Fairness

Quy Lam* Ketki Sheth†

Abstract

Deterrence theory assumes that harsher penalties reduce misconduct by raising its cost. Yet enforcement often depends on individuals who witness or respond to wrongdoing, and their behavior may also be shaped by the severity of punishment. Using a high-powered online experiment framed in a workplace setting, we test whether increasing penalties reduces willingness to report. We find that harsher penalties suppress reporting: employees are 1 percentage point less likely to report for every 10 percentage point increase in the penalty. This effect is mediated by perceived fairness: reporting declines only when penalties are seen as excessive. Despite the decrease in reporting, higher penalties still deter misconduct overall, as their direct deterrent effect outweighs the indirect increase in law-breaking caused by lower reporting rates. We also find that individuals support harsher penalties before a violation occurs, when deterrence is relevant, but prefer more lenient consequences after the offense has happened. This helps explain how policies can arise and persist in equilibrium, even when people are unwilling to enforce them in practice. These findings advance understanding of how social norms and perceptions of fairness shape the decisions of those who observe misconduct, and highlight the need to design sanctions that align with public willingness to act on violations. More severe penalties can discourage reporting, weaken enforcement, and ultimately undermine the effectiveness of deterrence mechanisms.

*University of California, Merced.

†University of Tennessee (Corresponding Author: ksheth2@utk.edu).

This study was approved by the University of California Merced IRB. This study was preregistered at the AEA RCT Registry (AEARCTR-0013393).

1 Introduction

Penalties are commonly used in law, workplaces, and schools to shift norms and deter misconduct. According to traditional deterrence theory, harsher penalties should reduce wrongdoing by making it more costly (Becker, 1968). But whether a penalty is enforced often depends on the willingness of others - such as witnesses, peers, and supervisors - to report violations.¹ Yet many witnesses hesitate to report wrongdoing (e.g., Drysdale, 2010, Brank et al., 2007, Brinkley and Saarnio, 2006). One explanation is that the severity of the punishment may shape whether observers choose to act. As punishments grow harsher, individuals may become more reluctant to report, perhaps due to concerns about disproportionate consequences or about wrongly accusing someone. This could reduce reporting and weaken enforcement, thereby undermining deterrence. Alternatively, more severe penalties may encourage reporting by signaling the seriousness of the offense, or may have no effect at all if individuals see their roles as simply enforcing established rules.

Understanding whether and how penalty severity affects reporting can help explain why enforcement is often uneven, why penalties do not escalate indefinitely, and why stronger sanctions sometimes fail to deter. This study examines the willingness of witnesses to report wrongdoing, focusing on corporate misconduct, where whistle-blowers and bystanders are often the only source of detection. We address three central questions: How does penalty severity influence reporting behavior? How does the decision to offend respond to the perceived likelihood of being reported? And what mechanisms explain changes in reporting?

To investigate these questions, we conduct a high-powered online lab experiment, framed in a workplace context, in which enforcement depends on employees' willingness to report misconduct by their managers. Managers can choose to "break a law" that benefits the firm at the expense of others, reflecting common real-world trade-offs between organizational gain and societal harm. Employees can report their manager for breaking the law, and we randomly vary the penalty the manager will face if reported: 5%, 40%, or 100% of their earnings. If employees do not report the violation, the manager is not penalized. This allows us to estimate how the likelihood of reporting misbehavior causally changes with respect to the severity of punishment. We also examine how managers respond to these varying penalty levels, both when they have no information about employees' reporting rates and when they receive information that serves as a proxy for the likelihood of being reported. This comparison enables us to isolate the indirect deterrent

¹For example, Dyck et al. (2010) analyzed 216 securities class action lawsuits filed against large U.S. corporations and found that only about 19% of them were brought forward by an employee, despite employees having significant access to internal information. Outside of corporate settings, observers are frequently present in episodes of misconduct. Philpot et al. (2020) find that in public conflicts across the UK, Netherlands, and South Africa, third parties were present in nearly all cases. Similarly, uninvolved witnesses are commonly documented in the majority of assaults, robberies, and other cases of criminal victimization (Hamby et al., 2016; Hart & Miethe, 2008).

effect operating through changes in reporting (i.e., getting caught).

We find that subjects are responsive to penalties: employees are 1 percentage point less likely to report a law-breaking manager for every 10 percentage point increase in the manager’s penalty. This suppression effect closely mirrors employees’ own beliefs, elicited through an incentive-compatible method. These participants accurately predicted that reporting would decline with higher penalties and provided estimates that were close in size to the observed effect. This alignment reinforces the credibility of our experimental estimates, demonstrating that those making reporting decisions anticipated how others would respond, suggesting that their intuition and experience reflect real-world patterns captured by our experiment. Furthermore, perceptions of fairness emerge as a key mechanism behind this response: reduced reporting occurs only among employees who stated that an appropriate penalty would be lower than the one imposed in the experiment. To confirm the robustness and generality of these findings, we replicate the core patterns in a separate set of hypothetical scenarios. We use both across-subject variation, where different participants are randomly assigned different penalty levels, and within-subject variation, where each participant evaluates multiple penalties for the same scenario. We find that reporting declines with harsher punishments, and this pattern is tied to when the punishment is perceived as excessive. In contrast, we do not find altruism, reciprocity, or law-abiding preferences explain the decline in reporting.

We also find that individuals support harsher penalties *ex ante* (before the offense is committed), but prefer more lenient sanctions *ex post* (after the misconduct has occurred). Participants randomly assigned to set a penalty before knowing whether an offense had occurred imposed substantially higher fines than those assigned to impose a penalty on an offender who had committed a violation. This comparison highlights the role of deterrence: when misconduct is still hypothetical, individuals appear more willing to endorse severe penalties, but once the violation has occurred, they often hesitate to enforce them and instead favor milder consequences. This gap helps explain why deterrence-focused penalties may exceed what people ultimately deem appropriate, and why such penalties may go unenforced in practice. The dynamic offers an explanation for how severe punishments can arise and persist in equilibrium, even when individuals are reluctant to uphold them.

Managers’ behavior also responds to changes in expected reporting: for every 10pp increase in employee reporting, managers are 6.7pp less likely to break the law. Although higher penalties suppress reporting, this effect is not large enough to eliminate the overall deterrent effect. The *direct* deterrent effect from harsher penalties outweighs the *indirect* increase in misconduct caused by reduced reporting rates, even when managers are informed about the probability of being reported. When managers lack information about the probability of being reported, the deterrent effect is even stronger, suggesting that managers incorrectly assume that higher penalties will increase the likelihood of being re-

ported. Indeed, their stated beliefs, elicited in an incentive-compatible method, indicate that they expect little change, and perhaps even an increase, in the likelihood of being reported as penalties rise. Such beliefs would make harsher sanctions even more effective as deterrents.

By using a laboratory experiment, we overcome key limitations that make it difficult to identify the causal effect of punishment severity on reporting behavior in observational settings. Natural variation in policy is rarely exogenous and often coincides with broader shifts—such as rising intolerance for corporate wrongdoing—that independently affect willingness to report. Even when variation is plausibly exogenous, higher penalties reduce misconduct, leaving less wrongdoing to observe and biasing the relationship between punishment severity and reporting. Moreover, enforcement typically involves multiple actors, and individuals often condition their decisions on what they expect others to do. If beliefs about how others enforce rules vary with penalty severity, they can confound the relationship between punishment and reporting. Our experiment overcomes these challenges by exogenously varying penalties while holding constant both the underlying misconduct and the actions of others. It also allows us to isolate mechanisms rarely observable in the field: we remove ambiguity about guilt and directly measure beliefs about the fairness of penalties. Unlike much prior lab work, our design involves real stakes, reducing concerns about hypothetical bias. Together, these features allow us to identify not just whether penalties deter, but also the behavioral channels through which they influence reporting and compliance.

We contribute to a longstanding, primarily theoretical, literature that has raised concerns about whether harsher punishments may paradoxically weaken enforcement by affecting the behaviors of actors across the enforcement chain.² Most empirical work to date focuses on courtroom actors such as prosecutors, judges, and juries, often in cases involving serious violent crimes. Devine (2012)’s review of the literature on penalty severity and jury behavior concludes that experimental lab studies generally find no reliable effect of penalty severity and that their external validity is typically limited to homicide cases.³ A more recent lab experiment with university students and real incentives eliminated ambiguity about guilt, a common feature in earlier studies, and found that harsher penalties reduced witness reporting, though the study was limited by a small sample size ($n = 39$) and did not investigate underlying mechanisms (Aimone et al., 2025). A handful of natural experiments have attempted to isolate the causal effect of penalty severity while holding offense severity constant, using policy shifts such as minimum sentencing laws or

²This idea has roots in early legal and philosophical thought and appears across several literatures under different labels. For example, Smith (1978) reflects on the risks of excessive sanctions, Kerr (1978) formalizes the concept as a severity leniency bias, and Kahan (2000) describes how “sticky norms” can lead well-intentioned laws to backfire when they conflict with prevailing moral intuitions.

³Examples of the lab experiments reviewed by Devine (2012) are generally lab experiments in which university students are mock jurors making hypothetical decisions, and treatment arms are often confounded by charge seriousness and allow for ambiguity of guilt (e.g., Vidmar, 1972, Kerr, 1978).

other historical changes. These studies suggest that harsher punishments can reduce the charges that prosecutors levy (Bjerk, 2005), increase plea bargaining (Lacasse & Payne, 1999), and lower jury conviction rates (Bindler & Hjalmarsson, 2018).⁴ While these natural experiments provide support that harsher penalties can reduce enforcement, they have not extended this evidence to witnesses, and by design, cannot identify underlying mechanisms. In particular, it remains unclear whether reluctance stems from concern for the wrongdoer and fairness, fear of false conviction, or beliefs about how others will respond. To date, the existing literature offers suggestive support that excessive penalties may reduce enforcement, but this evidence is drawn from a narrow set of contexts and actors, and the mechanisms underlying the effect remain largely unexplored.

Our study complements this body of work by providing causal evidence from a high-powered lab experiment that isolates the effect of punishment severity on a less-studied but critical form of enforcement behavior: witness reporting. Beyond improved identification through random assignment and incentivized decisions in the lab, we make several contributions to the literature. First, our design enables us to identify mechanisms that are difficult to disentangle in natural settings. By removing any ambiguity about guilt, we directly address one of the most prominent explanations for this theorized effect: fear of wrongful conviction. Our findings rule out potential wrongful conviction as a necessary condition and underscore the relevance of alternative explanations. The design also eliminates the influence of expectations about how others in the enforcement chain will behave, allowing us to isolate a direct response to the punishment the wrongdoer would face. We find that reduced reporting is concentrated among individuals who perceive the penalty as too harsh, suggesting that fairness concerns drive this behavior. This pattern emerges consistently across both incentivized and hypothetical scenarios and using both within- and across-subject variation.

A second contribution of our design is the ability to identify both how penalty severity affects reporting behavior and misconduct, including how changes in reporting, in turn, influence misconduct. To our knowledge, no prior study has measured these effects in tandem. This joint identification is essential, as the central concern is that diminished reporting undermines enforcement, potentially reversing the intended deterrent effects of harsher penalties and increasing violations. We also compare managers' actual behavior to their stated beliefs, which provides insight into whether potential wrongdoers correctly anticipate changes in enforcement, enhancing our understanding of how expectations about being reported interact with the deterrent effect of penalties.

⁴To our knowledge, there is only one natural experiment that touches on witness reporting, conducted in the context of whether mandatory arrest laws for domestic violence reduced reporting, but the findings remain inconclusive due to disagreements over how the analysis should be conducted. In Iyengar (2009)'s original analysis, she finds that mandatory arrest laws counterintuitively increased intimate partner homicides, driven by a reduction in the partners' willingness to report. However, more recent analysis that reviews this work by Chin and Cunningham (2019) finds errors in coding of the original laws and analysis, and finds no evidence that mandatory arrest laws increased homicides.

Third, our findings help explain why excessively harsh penalties may arise and persist in equilibrium, even when people are reluctant to enforce them. By showing that individuals tend to support stricter penalties when deterrence is relevant but prefer more lenient penalties once misconduct has occurred, we provide an explanation for why laws or rules that conflict with fairness norms are nonetheless enacted or maintained. This builds on a growing literature examining the implications of social norms. For example, Acemoglu and Jackson (2017) predict that when laws outpace social norms, individuals anticipate weak enforcement and become more likely to violate the law, potentially exacerbating misconduct. We provide empirical support for this prediction by showing that individuals are less likely to report wrongdoing when the punishment exceeds what they believe is socially appropriate.⁵ These findings also help reconcile puzzling patterns in the literature, such as why public support for severe punishments, like the death penalty, may coexist with reluctance to impose them in practice (Bindler & Hjalmarsson, 2018).

Finally, our paper expands the empirical and institutional context in which the causal effect of punishment severity has generally been studied. We show that reluctance to enforce harsh penalties is not limited to courtroom actors such as judges or jurors, but extends to bystanders and witnesses, broadening the scope of actors implicated in enforcement decisions that respond to penalty severity. To our knowledge, only one natural experiment has examined the effect of penalties on witness behavior, with the most recent reanalysis finding no evidence of reduced reporting (Chin & Cunningham, 2019), and one small-scale lab study has documented a decline in reporting (Aimone et al., 2025). Our study also moves beyond the context of violent crime by examining these dynamics in a workplace setting, where enforcement often depends on peer reporting.

The rest of the paper proceeds as follows. Section 2 describes the experimental design, Section 3 details our estimating strategy and pre-analysis plan, Section 4 presents our findings, and Section 5 concludes.

2 Experimental Design

2.1 Overview

We design a framed online lab experiment that simulates workplace decision-making, where employees may witness managerial misconduct and must decide whether to report it. Most participants are assigned the role of firm "employees" and grouped into "firms," while a small minority are designated as the "public." Participants were recruited through Prolific and completed the experiment remotely. They received a base participation pay-

⁵In contrast to our design, Acemoglu and Jackson (2017) emphasize a coordination-based mechanism in which enforcement weakens because individuals expect others not to uphold laws that exceed prevailing social norms.

ment of \$4, with additional earnings determined by their decisions in the experiment (converted at 100 points per \$1). Any penalties imposed during the experiment affected only these additional earnings, ensuring that no participant earned less than the guaranteed participation payment.

The Firm: For those assigned to be employees, the experiment consists of two interactive stages followed by a survey.⁶ In Stage 1 (Firm Identity Building), employees earn payouts based on their performance and that of their co-workers. In Stage 2 (Main Experiment), one participant per firm is randomly assigned the role of "manager." While earnings remain linked to the firm's performance, the manager can choose to "break the law," increasing firm members' earnings at the expense of the public. Employees then have the option to report a law-breaking manager, who incurs a penalty if reported. The severity of the penalty imposed on the manager is randomly assigned (5%, 40%, or 100% of the manager's earnings).⁷ This variation allows us to identify the causal effect of penalty severity on reporting behavior. The final stage consists of a survey designed to further explore participants' perceptions and motivations.

Comprehension questions and attention checks were included throughout, with participants removed from the experiment if they failed to answer correctly within two attempts. 1,718 individuals were assigned a penalty treatment, and 1,403 successfully passed all checks and completed the experiment. This non-completion is balanced across treatment arms (see Appendix Table A1). We do observe that those assigned to be a manager were more likely to not complete the experiment. Participants were grouped into firms of 4 to 6 members, resulting in 256 firms with 256 managers and 1,147 employees. The average experiment duration was 25 minutes, and participants earned an average of 4.83 USD, excluding the participation payment. The sections below describe each stage in greater detail, presented in the chronological order in which they were given to the subjects.^{8,9}

To ensure participants perceived misconduct and reporting with the psychological and social weight these decisions carry in real-world settings, the experiment employs meaningful language rather than neutral labels. This approach, commonly used in lab experiments studying such social issues as bribery and pollution, has been shown to increase the salience of moral and social costs, leading to behavior that better reflects natu-

⁶The structure of these first two stages largely follows the experiment used in Butler et al., 2020.

⁷All random assignments were based on a different computer-generated random number at the relevant stage of the experiment, with equal probability of assignment unless otherwise noted. The exception is that the manager role is selected with 20% probability.

⁸Grouping participants into firms occurred after the experiment to ensure that participants' choices influenced others as described, while avoiding delays caused by waiting for others to make decisions in real time. Subjects are therefore informed of their individual performance throughout the experiment, but informed of their final payment only after completing the experiment. Despite this procedural choice, all information provided to subjects was accurate and truthful, and no deception was used at any point in the experiment.

⁹Participants earned points in all three stages. Stage 1 or 2 was randomly selected for payment, and Stage 3 was always paid.

rally occurring decision-making (Alekseev et al., 2017).¹⁰ By using meaningful language, we improve the external validity of our findings, ensuring that participants experience reporting and misconduct in a way that mirrors their societal implications rather than treating them as abstract choices. A firm setting is especially well-suited for studying these dynamics, as compliance with laws and policies is frequently enforced through the reporting of misconduct by others.

The Public: As our primary focus is on employees and managers, we assigned only two groups of five participants (10 participants total) to the public role. Public members played no active role in the experiment, but were financially affected by managerial decisions. Each group of public members started with 3,730 points, losing 150 points for each firm that broke the law. If 25 or more firms engaged in misconduct, public members' final earnings were reduced to zero.

2.2 Stage 1: Firm Identity Building

In the first stage, subjects are told that in this experiment, some individuals are assigned to firms while others are assigned to the public. They are informed that their firm will consist of four to six members, that they will remain in the same firm for the entire experiment, and that five individuals have been assigned to represent the public for their session.

The goal of this stage was to build a sense of firm identity among participants. To achieve this, firm members completed a shared task with interdependent payoffs: each participant earned points based on both their own performance and that of their co-workers. Prior work demonstrates that such interdependent structures foster loyalty and identification with the group in laboratory settings.¹¹ We confirmed participants understood the interdependence of their payoffs through comprehension checks administered before the task. In the task itself, firm members had 60 seconds to decode words into numbers using a predefined key. For each correct word decoded by any firm member, all members earned 55 points. This payoff structure mimics real workplace environments, where economic outcomes are often shared and individual contributions are not always visible.

¹⁰In Alekseev et al., 2017 review of contextual language in lab experiments, they find that when evocative framing affects behavior in the lab, it generally does so in a way that improves alignment with real-world responses. In addition, they find that framing the experiment in a meaningful context, such as the workplace, often enhances comprehension and reduces the risk of participants imposing their own interpretations a more abstract context.

¹¹In a lab experiment that uses the same structure, (Butler et al., 2020) document that firm members contribute more in a public good game after such a firm identity building stage.

2.3 Stage 2: The Main Experiment

Subjects are told that their firm’s decisions in this stage may affect the earnings of participants assigned to the public. They are informed that one member of each firm will be randomly assigned to become the manager, with the remaining members acting as employees, and that how points are earned will differ by these roles. Employees will engage in a task in which they will earn 70 points for themselves and 35 points for the firm, referred to as the firm’s surplus, for each correct answer. The manager will earn a fixed salary of 1,500 points, and will engage in a harder task. Successful performance in the harder task (answering 8 of 10 questions correctly) will double the firm’s surplus. The firm’s surplus is then redistributed back to the members of the firm: managers receive 60% of the firm surplus and the remaining 40% is divided evenly among the remaining employees in the firm. After receiving this introductory information, participants are randomly assigned to either the manager or employee role, at which point their experience in the experiment diverge.

Managers: The manager is told they can attempt to double the firm’s surplus by correctly identifying the number of zeros in 8 out of 10 matrices (each 10 by 5) within 60 seconds. Alternatively, they can choose to “break the law,” which automatically doubles the surplus but reduces each public member’s earnings by 150 points. Managers are informed that employees may report this misconduct and earn 10 points for doing so. One employee is randomly selected to determine whether the manager is penalized. If that employee reports the manager, the manager’s experimental earnings are reduced by a randomly assigned penalty of 5%, 40%, or 100%, our key independent variable¹². If the selected employee does not report, no penalty is applied.

To ensure managers understand their earnings structure and choices, they are required to answer comprehension questions before making any decisions. These questions cover payoffs, the consequences of breaking the law (to the firm and the public), and the penalty rates they would face both if reported and if not reported. Only managers who correctly answered all questions within two attempts are included in the analysis sample, ensuring full understanding of the key experimental variable.

The manager then decides whether or not to break the law. After recording this decision, the manager is told their choice to break the law can depend on the likelihood of being reported by an employee. Specifically, the manager is presented with 10 scenarios, each representing a 10pp range in reporting likelihoods, and are asked whether or not they want to break the law for each scenario. For example, “If the probability of being reported is between 31 and 40%, what do you want to do? Break law or continue to task?” They are informed that “we will use your decision in the scenario that matches the actual probability of being reported by an employee (the actual probability is based on

¹²Note that this variable is randomly assigned at the individual level as the firm groupings are completed after the experiment.

the percent of employees that reported law-breaking managers in the previous sessions of this experiment).” The reporting rate of previous sessions is an effective proxy for the likelihood of being reported in the current session. We refer to the first decision made by the manager as “Unconditional” and the decisions based on the ten scenarios as “Conditional”.

We then ask the manager to estimate the proportion of employees who reported on their manager in the previous day’s session. To encourage accuracy, we use an incentive-compatible belief elicitation method (Danz et al., 2020, Burdea and Woon, 2022): managers receive 10 points if their estimate falls within ± 5 percentage points of the true proportion. After submitting their estimate, we implement the conditional decision that corresponds to the actual reporting rate from the prior session. Managers are not told which scenario was selected. If their decision at that reporting level was not to break the law, they instead complete the counting task.

Employees:

After the introduction, those selected to be employees are first shown instructions about their task: counting the number of zeros in a 10 by 3 matrix (10 rows and 3 columns) within one minute. Before beginning the task, they answer comprehension questions about how much they and the firm will receive. Only those who answer correctly within two attempts are included in the analysis sample, ensuring that all employees understand the payoff structure.

After completing the number counting task, the employee is made aware of the choice that had been provided to the manager: to attempt a task to double the firm surplus or to “break the law” which automatically doubled the surplus but reduced earnings of each member of the public by 150 points. We then ask the employee what they believe the appropriate penalty rate should be for a manager who broke the law (i.e., the percent of the manager’s earnings that should be reduced) (*Apprpt Pnlty*).

Next, employees are informed of several key points: they can choose to report their manager for breaking the law; the specific penalty the manager will face if reported; that no penalty is applied if the manager is not reported; that reporting earns the employee ten additional points; and that the public is neither informed of nor benefits from reporting. They are also told that one employee’s decision will be randomly selected for implementation. To ensure these key details are made salient and are well understood, employees must answer comprehension questions confirming their awareness of the penalty amount, the fact that the firm’s surplus increases when the manager breaks the law, and that the public loses 150 points as a result. They must also specify the exact penalty rate the manager would face if reported and if not. Only employees who answer all questions correctly within two attempts are included in the analysis sample.

The employee then decides whether to report their manager for breaking the law (*Report*). To measure employees’ willingness to report, we use the strategy method: the

employee is asked to make a reporting decision without knowing whether their manager actually broke the law. This approach allows us to assess their willingness to report even in the absence of misconduct. The key advantage of this method is that it eliminates selection bias based on which managers actually break the law and helps preserve our sample size. Studies have shown that this strategy method produces responses comparable to those made when employees observe the manager’s actual decision to break the law, ensuring that the method has external validity to the real-world context where employees typically decide whether to report after witnessing misconduct (Brandts and Charness, 2000, Brandts and Charness, 2011).

2.4 Stage 3: Survey Questions

The final part of the experiment is the same for both employees and managers, and consists of a series of questions to further explore how the severity of penalties influences the decision to report.

Priors in the experiment: Subjects are informed of the proportion of employees that reported the manager in the previous sessions for firms that faced the same penalty fine that they had experienced.

We then elicit participants’ beliefs about two outcomes at each penalty level (5%, 40%, and 100%): (1) the proportion of employees who reported law-breaking managers, and (2) the proportion of managers who chose to break the law. These beliefs are elicited using an incentive-compatible frequency method (see Danz et al., 2020, Burdea and Woon, 2022). Specifically, participants are told that we will randomly select 10 past participants who held a given role (employee or manager) in a firm assigned to one of the three penalty rates. We will then randomly select one of the subject’s six guesses, and if their selected guess matches the actual number of participants who behaved accordingly, they receive a bonus of 200 points. To avoid priming participants to think about managers’ behavior while forming beliefs about employee reporting, we randomly assigned half of the sample to report both sets of beliefs simultaneously, and the other half to report beliefs sequentially (first about employee reporting, and then about manager law-breaking).

Hypothetical scenarios and penalty severity: We then present four hypothetical scenarios involving clear rule violations and varying levels of punishment, and ask participants whether they would report the violator. These scenarios are explicitly framed to mirror key features of our main experiment: participants are told that the behavior is unambiguously in violation of a law or rule, that reporting is completely anonymous, and that the violator will only be caught if the participant chooses to report. In the latter two scenarios, we also examine whether the perceived appropriateness of the punishment influences reporting behavior.

First, we describe a scenario where a colleague engages in discriminatory behavior

and we randomly assign the severity of the consequence if the subject reported their colleague. The randomly assigned severity is one of the following four consequences: formally warned about their behavior, formally warned about their behavior and monitored in future decisions on promotions, suspended from their job without pay, and fired. The subject states whether they would report their colleague given the randomly assigned consequence.

Second, we describe a scenario in which the subject is a student and a classmate cheats on an exam. We provide the subject a list of six increasing consequences if the classmate is reported: verbal warning, reduced grade on the exam, zero on the exam, fail the course, fail the course and be suspended from the institution, and expelled (i.e., permanently removed) from the institution. We ask the subject whether they would report their classmate for each of the given consequences. These first two scenarios both capture whether the severity of consequences impacts the likelihood of reporting, with the former using variation across subjects and the latter using variation within subjects.

Third, we describe a scenario where the subject witnesses graffiti being spray-painted on a public park wall. We randomly assign the severity of the consequence if the subject reported the person from one of the following: responsible to pay for the damages (estimated at \$200) only, responsible to pay for the damages (estimated at \$200) plus 100 hours of community service, and responsible to pay for the damages (estimated at \$200) plus 100 hours of community service and serving 2 months in prison for felony vandalism¹³. In addition to asking whether the subject would report the vandalizer, we also ask them if the consequence for the vandalism is "excessive (that is, too severe)".

In our fourth scenario, we describe a person violating the American Disabilities Act by illegally parking in an Accessible Parking Space. Subjects are presented with a series of escalating fines for the violation— \$100, \$500, \$1,000, and \$2,000—and are asked two questions for each fine: whether they consider the fine excessive and whether they would report the violator. These latter two scenarios build on the analysis by exploring how the perceived appropriateness of the penalty influences the decision to report, with the third scenario using variation across subjects and the fourth using variation within subjects.

Penalty Preferences: Pre- and Post-Violation Comparisons:

A central insight of our study is that the deterrent effect of penalties may come into conflict with individuals' willingness to enforce them. If people care about the consequences imposed on wrongdoers, they may support harsher penalties in principle, but resist applying them once a violation has occurred. If this is true, the consequence considered appropriate before misconduct ('ExAnte') should be greater than the consequence deemed appropriate after it occurs ('ExPost'). This difference arises because, before the act, the penalty serves as both a deterrent and punishment, whereas after the misconduct

¹³For this last option, there was a typo in the experiment so that the last part read as "2 months of imprisonments for felony vandalism."

has occurred, deterrence is no longer relevant – only the punishment remains.

We test this idea directly using a randomized, between-subjects design. Participants are asked to set a fine for littering in a protected area, but randomly assigned to one of two conditions. In the *ExAnte* condition, participants are told: "Your advisors have told you that fines for such a violation in other countries typically range from \$500 to \$4000. What would you determine the fine to be?" In the *ExPost* condition, participants are instead told: "You are determining the fine for a person that is guilty of littering in a protected area. The law advises that the fine should range between \$500 to \$4000. What fine will you impose on the person"? This design uses across-subject variation to examine whether the fine imposed *ExPost* is lower than the one set *ExAnte*, suggesting that, in the absence of a deterrent effect, subjects prefer violators to face lower penalties.

This distinction is critical because it helps explain why penalties set with deterrence in mind may ultimately be perceived as too harsh when individuals must decide whether to enforce them through reporting. While individuals may initially support strict penalties as a deterrent, they may hesitate to impose them in practice, contributing to under-reporting of misconduct. This gap between ex-ante penalty preferences and ex-post reluctance to impose harsh consequences highlights a fundamental challenge in enforcement: penalties designed for deterrence may exceed what people actually feel comfortable imposing, potentially undermining the effectiveness of reporting mechanisms and law enforcement.

Preferences: In the final questions of the experiment, we elicit participant preferences on law-abiding, altruism, positive reciprocity, and negative reciprocity. Following the 2023 British Social Attitudes (BSA) survey (Research, 2023), we measure individuals' willingness to disobey laws they perceive as unjust by asking respondents to rate, on a 5-point Likert scale, their agreement with the statement: "The law should always be obeyed, even if a particular law is wrong." A value of 1 indicates strong agreement, while higher values reflect greater disagreement (*Law-Abiding Reluctance*). To measure altruism and reciprocity, we use questions from the shorter module in Falk et al., 2023, which have been tested and validated across multiple countries.¹⁴ Since both misconduct and reporting behavior can be influenced by individual personality traits, we include these questions as

¹⁴For each preference, we construct an index by standardizing the relevant variables and taking their average. We ask 2 questions regarding to altruism: 1) *Altruism, Self-Reported*: How willing are you to give to good causes without expecting anything in return?; 2) *Altruism, Donation*: Imagine the following situation: Today you unexpectedly received \$1,000 . How much of this amount would you donate to a good cause? (Values between 0 and 1,000 are allowed); We ask 2 questions regarding positive reciprocity: 1) *Pos. Rec., Self-Reported*: When someone does me a favor. I am willing to return it.; 2) *Pos. Rec., Gift*: A stranger helps you. Do you give one of the presents to the stranger as a "thank-you"-gift? If so, which present do you give to the stranger?; We ask 3 questions regarding negative reciprocity: 1) *Neg. Rec., Self*: How willing are you to punish someone who treats you unfairly, even if there may be costs for you?; 2) *Neg. Rec., Other*: How willing are you to punish someone who treats others unfairly, even if there may be costs for you?; 3) *Neg. Rec., Revenge*: If I am treated very unjustly, I will take revenge at the first occasion, even if there is a cost to do so.

controls and to explore potential mechanisms that may mediate the effect of penalties on misconduct and reporting decisions.

2.5 Summary Statistics

Table 1 reports summary statistics for our sample. The first three rows present age, gender, and race—baseline respondent characteristics provided by Prolific at registration. The average respondent is 37 years old, with a roughly even gender split and approximately 40 percent identifying as non-white. The remaining rows summarize the preferences measured during the experiment. Columns 6 through 9 confirm that respondents are balanced on these characteristics across penalty treatment arms and between managers and employees.

Table 1: Summary Statistics on Participant Demographics and Preferences

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Employee	Manager	5% fine	40% fine	100% fine	Mngr. = Empl.	5% = 40% + 100%	40% = 5% + 100%	100% = 5% + 40%
Age	37.18	37.29	37.27	37.12	37.22	-0.11	-0.09	0.12	-0.02
Female	0.49	0.56	0.49	0.53	0.50	-0.07*	0.02	-0.03	0.01
Non-white	0.43	0.36	0.42	0.41	0.42	0.07*	-0.00	0.00	0.00
Altruism Index	-0.01	0.03	-0.03	0.01	0.02	-0.03	0.04	-0.01	-0.03
Altruism, Self-Reported	7.75	7.92	7.74	7.74	7.86	-0.17	0.07	0.05	-0.12
Altruism, Money Donation	135.14	134.43	128.66	139.28	137.39	0.71	9.63	-6.21	-3.64
Pos. Rec. Index	-0.01	0.04	-0.06	0.05	0.01	-0.05	0.09	-0.07	-0.02
Pos. Rec., Self-Reported	8.93	8.99	8.89	9.01	8.94	-0.06	0.08	-0.09	0.01
Pos. Rec., Gift	4.65	4.75	4.54	4.76	4.71	-0.10	0.19	-0.13	-0.07
Neg. Rec. Index	0.01	-0.04	-0.03	0.03	0.01	0.05	0.04	-0.04	-0.01
Neg. Rec., Self	4.99	4.69	4.81	5.02	4.98	0.30	0.19	-0.12	-0.07
Neg. Rec., Other	5.71	5.54	5.63	5.86	5.56	0.16	0.08	-0.27	0.18
Neg. Rec., Revenge	2.87	2.90	2.82	2.81	2.98	-0.04	0.08	0.10	-0.17
Law-Abiding Reluctance	3.01	3.01	3.07	3.01	2.95	-0.00	-0.09	0.00	0.09
Observations	1147	256	477	437	489	1403	1403	1403	1403

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Age, Female and Non-white are characteristics that were measured prior to the experiment during Prolific registration. The remaining preferences were measured in Stage 3 of the experiment. For each preferences, we report the Index, followed by the underlying questions that form the Index. Columns 1 to 5 report the mean by treatment arm and firm role. Columns 6 to 9 report t-statistics from tests of equality between groups: Column 6 compares employees and managers; Columns 7 to 9 compare each treatment arm against the other two.

3 Estimating Strategy

The following is a summary of our registered pre-analysis plan.

3.1 Primary Outcomes for Employees

Our key question of interest is whether reporting is sensitive to the penalty imposed for breaking the law (i.e., $\hat{\gamma}_1 < 0$):

$$Report_{if} = \gamma_0 + \gamma_1 Fine_f + \epsilon_{if} \quad (1)$$

where *Report* is an indicator for whether employee i at firm f reported on a law-breaking manager. *Fine* is the randomly assigned penalty fine that the manager faces

at firm f . We test for heterogeneity in Eq (1) as a function of altruism, reciprocity, law-abiding, and an indicator for whether their reported appropriate penalty was below the imposed penalty (i.e., $1[AppropriatePenalty < Fine]$).¹⁵

3.2 Primary Outcomes for Managers

We test whether at higher probabilities of being reported, the manager is less likely to break the law (i.e., $\hat{\beta}_1 < 0$):

$$BreakLaw_{ifs} = \beta_0 + \beta_1 ProbabilityofReported_s + \epsilon_{ifs} \quad (2)$$

where *BreakLaw* is an indicator variable of whether manager i from firm f in scenario s decided to break the law. *ProbabilityofReported* corresponds to the 10 reporting scenarios, coded from 1 to 10, increasing in 10pp increments of the proportion of employees reporting law-breaking managers in previous sessions. We cluster standard errors at the manager.

We next test whether managers are less likely to break the law when faced with a higher penalty (i.e., $\hat{\lambda}_1 < 0$):

$$BreakLaw_{ifs} = \lambda_0 + \lambda_1 Fine_f + \epsilon_{ifs} \quad (3)$$

where *Fine* is the penalty fine that manager i is facing at from firm f for all scenario s . Standard errors are clustered at the manager.

We expand the model to test for how the penalty interacts with the probability of being reported, expecting that at higher penalties, the probability of being reported reduces the probability of breaking the law more so (i.e., $\hat{\lambda}_2 < 0$):

$$BreakLaw_{ifs} = \lambda_0 + \lambda_1 Fine_f + \lambda_2 Fine_f \cdot ProbabilityofReporting_s + \lambda_3 ProbabilityofReported_s + \epsilon_{ifs} \quad (4)$$

We can estimate the expected reduction in crime holding reporting constant using Eq. (3): $\Delta BreakLaw = \hat{\lambda}_1 \Delta Fine$. This will overestimate the reduction in crime by Eq. (2): $\Delta BreakLawProb = \hat{\beta}_1 \Delta ProbabilityofReported$, where $\Delta ProbabilityofReported = \hat{\gamma}_1 * Fine$.

Finally, we test for the effect of the penalty on breaking the law with the manager having information on reporting. That is, we limit observations to the scenario that was actually implemented.

¹⁵For preferences measured by multiple variables, we construct the z-score for each variable and then average these to create an index for each preference.

$$BreakLawConditional_{if} = \delta_0 + \lambda_1 Fine_f + \epsilon_{if} \quad (5)$$

where observations are limited to the scenario that was implemented.

3.3 Secondary Outcome: Unconditional Law Breaking by the Manager

The previous estimations are based on the condition that managers have a proxy for information on the probability of being reported. But we can also directly estimate the effect of the penalty on the decision to break the law without this information.

$$BreakLawUncond_{if} = \eta_0 + \eta_1 Fine_f + \epsilon_{if} \quad (6)$$

where *BreakLawUncond* is an indicator variable for if manager *i* from firm *f* decided to break the law when no information or conditioning on previous reporting was provided. In addition for testing for statistical significance relative to the penalty having no effect, we also compare η_1 to λ_1 .

3.4 Secondary Outcomes: Beliefs on reporting and law-breaking

We compare the results in the experiment to subjects' priors on how penalties affected the manager's decision to break the law and employees' decision to report law-breakers:

$$Y_{ip} = \delta_0^{Prior} + \delta_1^{Prior} PriorFine_{ip} + \epsilon_{ip} \quad (7)$$

where *Y* is either *PriorBreakLaw*, subject *i*'s belief of the number of managers (out of 10) that broke the law when faced with penalty *p*, or *PriorReport*, subject *i*'s belief of the number of employees that reported their manager when faced with penalty *p*. We estimate Eq.7 for both managers and employees separately. For beliefs on breaking the law, we compare the magnitude of δ_1^{Prior} to δ_1 (the effect of the penalty on breaking the law with perfect information on reporting), to η_1 (the effect of the penalty on breaking the law without perfect information on reporting), and to λ_1 (the effect of the penalty on breaking the law with perfect information on reporting for all levels of reporting). For beliefs on reporting, we compare the magnitude of δ_1^{Prior} to γ_1 (the effect of the penalty on reporting in the experiment).

4 Results

4.1 Reporting as a Function of the Penalty

Table 2 presents the main results of our study, showing that higher penalties reduce the likelihood of reporting. In the Main Experiment, employees were significantly less likely to report a law-breaking manager when the randomly assigned fine was higher. A 10pp increase in the penalty rate led to a 1pp decrease in reporting (Column 1), a result that is statistically significant at the 1% level. This effect is especially pronounced when the penalty is most severe: employees assigned to the 100% penalty condition were 9pp less likely to report than those assigned to the 5% condition, an 11% reduction in reporting (Column 2). These results are robust to the inclusion of day fixed effects (Column 3), subject demographic controls (Column 4), and subject preferences and their belief of the appropriate penalty (Column 5 and 6).¹⁶

Table 2: Punishment Severity and Reporting Rates

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Reported	Reported	Reported	Reported	Reported	Reported	Prior Report	Prior Report
Fine	-0.0981*** (0.0329)		-0.0891*** (0.0332)	-0.0856** (0.0335)	-0.0872*** (0.0336)	-0.0886*** (0.0323)		
40% fine		-0.0106 (0.0309)						
100% fine		-0.0893*** (0.0314)						
Prior Fine							-0.0805*** (0.0145)	
Prior 40% fine								-0.0232*** (0.00844)
Prior 100% fine								-0.0757*** (0.0141)
Day Fixed Effects	No	No	Yes	Yes	Yes	Yes	Yes	Yes
Demographics	No	No	No	Yes	Yes	Yes	No	No
Preferences	No	No	No	No	Yes	Yes	No	No
Apprt. Penalty	No	No	No	No	No	Yes	No	No
Observations	1147	1147	1147	1132	1129	1128	3429	3429
Mean Reported (Fine 5%)	0.777	0.777	0.777	0.777	0.777	0.777		
Mean Prior Report (Fine 5%)							0.600	0.600

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Reported is a binary variable equal to 1 if the employee chose to report their manager in the Main Experiment. Prior Report is the employee's belief about the proportion of participants in prior sessions who reported a law-breaking manager. Mean Reported (Fine 5%) is the mean reporting rate for employees assigned to the 5% treatment arm. Mean Prior Report (Fine 5%) is the average belief employees held about the reporting rate under the 5% penalty condition. Fine and Prior Fine indicate the randomly assigned penalty level—5, 40, or 100 percent of the manager's earnings. (Prior) 40% Fine and 100% Fine are indicators for the respective fine rate. Demographics are Age, Female, and Non-White. Preferences are Altruism Index, Positive Reciprocity Index, Negative Reciprocity Index, and Law Abiding Reluctance. Appr. Penalty refers to the penalty level the employee reported as appropriate for a manager who breaks the law. Columns 1–6 include one observation per employee. Columns 7–8 include three observations per employee, one for each fine level. Columns 7–8 include an indicator for whether priors were elicited simultaneously. Standard errors are robust in Columns 1 and 5, and clustered at the individual level in Columns 7–8.

Importantly, reporting behavior is systematically related to several theoretically rel-

¹⁶Appendix Table A2 reports all independent variables used in the regressions for Table 2

evant predictors (see Appendix Table A2). Participants who believed higher penalties was appropriate when directly asked what consequence a law-breaking manager should face were more likely to report the misconduct. Similarly, those who expressed greater willingness to report in hypothetical scenarios were also more likely to report in the Main Experiment, demonstrating consistency between stated intentions and incentivized behavior. Law-abiding attitudes were also predictive: individuals who indicated stronger commitments to obeying laws, even when perceived as unjust, were more likely to report their manager’s wrongdoing. These patterns underscore the internal coherence of subjects’ responses and lend credibility to the behavioral variation observed in our main results. In contrast, measures of altruism, positive reciprocity, and negative reciprocity were not significant predictors of reporting, highlighting the distinctive influence of perceived punishment fairness and normative views about legal compliance.

This behavior aligned with employees’ priors regarding the relationship between penalty severity and the likelihood of being reported. Specifically, employees believed that higher fines would reduce the likelihood of reporting (Table 2, Column 7). They expected that a 10pp increase in the fine would reduce reporting by 0.8pp, a prediction that is very close in magnitude and not statistically different from our causal experimental estimates. When asked to estimate reporting rates at each specific penalty level, employees’ predictions closely tracked both the direction and the size of the observed treatment effects: reporting declined as penalties increased, and the magnitudes of the predicted percentage point declines were not statistically different from the actual decline observed in the experiment (Table 2, Column 8). While subjects’ priors slightly underestimated the overall level of reporting (60% vs 77%), they were highly accurate in predicting the differences in reporting rates across varying penalty levels. The close alignment between employees’ predictions and the experimental results reinforces the credibility of our causal estimates, showing that participants who faced the reporting decision themselves could anticipate how others would respond to varying penalty severities. This suggests the observed effects reflect intuitive and socially understood dynamics, rather than artifacts of the experimental design.

This pattern is also reflected in our hypothetical survey questions on reporting misconduct, which incorporate both within- and across-subject variation. In Table 3, Column 1, we find that subjects randomly assigned to a more severe penalty for a colleague who discriminates are statistically less likely to report the behavior. Similarly, subjects randomly assigned to be shown a more severe penalty for observing vandalism are also less likely to report it (Column 3). We find a similar relationship on subjects’ willingness to report on a cheating classmate or a parking violation using within subject variation (Table 3, Column 2 and 4): subjects state they are less likely to report as penalties become more severe. When primed to consider whether the penalties are excessive, we continue to observe that subjects are less likely to report vandalism (Column 5, across-subject

variation) and parking violations (Column 6, within-subject variation) as penalties become more severe. Notably, when penalties are not perceived as excessive, the negative relationship between severity and reporting disappears. This suggests that perceptions of fairness, rather than severity alone, drive the decline in willingness to report. Appendix Table A3 confirms the robustness of these results using indicators for each specific penalty level rather than an ordinal scale, showing that reporting rates consistently decline with more severe penalties in within-subject designs and generally exhibit the same pattern in between-subject designs.

Table 3: Hypothetical Reporting Scenarios

	(1) Reported	(2) Reported	(3) Reported	(4) Reported	(5) Reported	(6) Reported
Penalty Discrimination	-0.0270*** (0.00871)					
Penalty Cheating		-0.124*** (0.00317)				
Penalty Vandalism			-0.138*** (0.0156)		0.00925 (0.0186)	
Penalty Illegal Parking				-0.172*** (0.00477)		-0.00256 (0.00850)
Vandalism Fine: 1[Excessive]					-0.404*** (0.0808)	
Penalty Vandalism*1[Excessive]					-0.0551* (0.0333)	
Penalty Illegal Parking: 1[Excessive]						-0.538*** (0.0323)
Penalty Illegal Parking*1[Excessive]						-0.0255** (0.0120)
Observations	1399	8412	1401	5611	1401	5610
Mean Reported (Least Severe)	0.874	0.749	0.708	0.764	0.708	0.764

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Reported is a binary variable equal to 1 if the subject chose to report the violator for the corresponding penalty in the scenario. Mean Reported (Least Severe) is the average reporting rate under the least severe penalty. Penalty is an ordinal variable that increases with the severity of the punishment shown in the scenario. 1[Excessive] is an indicator equal to 1 if the subject judged the given penalty to be excessive. Observations equal the number of subjects multiplied by the number of penalties shown per scenario. Standard errors are clustered at the individual level.

4.1.1 What Drives This Relationship

A key factor driving the relationship between reporting behavior and penalty severity is the subject's perception of whether the fine is appropriate. As shown in Table 4, Column 1, we find that the entire relationship is concentrated among individuals who perceive the fine to be higher than what they believe is warranted. While lacking statistical precision, this relationship does not appear to be limited to only the most extreme penalty (Column 2). Among those for whom the fine was lower than their stated view of what was appropriate, there appears to be no relationship between the severity of the fine and the likelihood of reporting the manager. This suggests that employees are more likely to report violations when the fine is perceived as fair, and that higher penalties need not suppress enforcement if they align with prevailing norms of appropriateness.

As noted earlier, this pattern is mirrored in the hypothetical scenarios: subjects are significantly less likely to report when they view the punishment as excessive, and the link between penalty severity and reporting disappears when the consequence is viewed as reasonable (Table 3 and Appendix Table A3, Column 5 and 6). These results underscore the central role of perceived fairness in shaping enforcement behavior, highlighting that penalties viewed as too harsh can backfire by deterring individuals from reporting misconduct.

Table 4: Punishment Severity and Reporting Rates Heterogeneity

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Reported	Reported	Reported	Reported	Reported	Reported	Suggested Fine	Suggested Fine
Fine	-0.0197 (0.0442)		-0.0976*** (0.0330)	-0.0966*** (0.0330)	-0.0989*** (0.0329)	0.00148 (0.0929)		
Apprt. Penalty	0.00310*** (0.000497)	0.00308*** (0.000496)						
Apprt. Penalty < Fine	-0.100** (0.0440)							
40% fine		0.0149 (0.0318)						
100% fine		-0.00649 (0.0405)						
Apprt. Penalty < 5% * 5% Fine		-0.154 (0.111)						
Apprt. Penalty < 40% * 40% Fine		-0.0844 (0.0565)						
Apprt. Penalty < 100% * 100% Fine		-0.113** (0.0485)						
Neg. Rec. Index			0.00616 (0.0244)					
Fine * Negative Reciprocity Index			-0.0308 (0.0400)					
Pos. Rec. Index				-0.0207 (0.0266)				
Fine * Positive Reciprocity Index				0.0484 (0.0428)				
Altruism Index					0.00939 (0.0244)			
Fine * Altruism Index					0.0138 (0.0428)			
Law-Abiding Reluctance						-0.00497 (0.0180)		
Fine * Law-Abiding Reluctance						-0.0336 (0.0296)		
Ex Post							-150.4*** (52.05)	-165.3*** (49.82)
Observations	1144	1144	1146	1146	1147	1145	1403	1284
Mean Reported (Fine 5%)	0.777	0.777	0.777	0.777	0.777	0.777		
Ex Ante Mean							1065.6	1065.6

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Reported is a binary variable equal to 1 if the employee chose to report their manager in the Main Experiment. Mean Reported (Fine 5%) is the mean reporting rate for employees assigned to the 5% treatment arm. Suggested Fine refers to the fine amount proposed by the participant in a hypothetical scenario, either before ex ante or after ex post the misconduct occurred. Ex Ante Mean is the average fine amount proposed prior to the misconduct. Fine indicate the randomly assigned penalty level—5, 40, or 100 percent of the manager's earnings. 40% Fine and 100% Fine are indicators for the respective fine rate. Appr. Penalty refers to the penalty level the employee reported as appropriate for a manager who breaks the law. Altruism Index, Positive Reciprocity Index, Negative Reciprocity Index, and Law Abiding Reluctance refer to elicited subject preferences. Ex Post is an indicator for whether the subject was randomly assigned to propose a penalty after the misconduct occurred. Columns 1–6 include one observation per employee. Columns 7 and 8 is one observation per subject. Column 8 limits observations to those who reported a suggested fine between 500 and 4000. Standard errors are robust.

Standard economic models of deterrence assume that individuals weigh the expected cost of punishment, which depends on both the severity of the penalty *and* the likelihood of detection. But this logic has generally assumed that enforcement is automatic. Our results show that when a penalty is perceived as unfair, individuals along the enforcement chain, such as witnesses, are less willing to report or punish wrongdoing. In this way, the effectiveness of deterrence depends not only on the costs faced by potential violators, but also on whether those responsible for enforcing the rules view the penalty as justified. These findings highlight the unintended consequences of imposing penalties that exceed

prevailing social norms of what is considered appropriate.

Unlike the perceived appropriateness of the fine, we continue to find no evidence that altruism, positive reciprocity, or negative reciprocity explain or mediate the relationship between penalty severity and the decision to report (Table 4, Columns 3 to 5).¹⁷ However, the pattern for law-abiding reluctance, measured as a lower willingness to follow laws perceived as unjust, suggests a potential moderating role (Column 6). When we interact this measure with the fine level, the largest effect appears on the interaction term, and the main effect of the fine becomes close to zero. While these estimates are not statistically precise, they suggest that the suppression of reporting under high penalties may be concentrated among individuals who express less willingness to obey laws unconditionally. This pattern aligns with the heterogeneity we observe in fairness perceptions: in both cases, individuals appear less willing to report wrongdoing when doing so conflicts with their personal sense of justice.

Our comparison of fines imposed before versus after misconduct occurs provides insight into *why* excessively high penalties may persist, even if individuals are reluctant to impose them on offenders. If subjects care about the consequences perpetrators face, the penalty deemed appropriate before the misconduct (*ex ante*) should exceed the penalty deemed appropriate after it occurs (*ex post*). This difference arises because *ex ante* penalties are designed to deter, but deterrence is no longer relevant once the act has occurred. To explore this dynamic, we asked subjects to imagine they were responsible for setting a fine for littering in a protected area, with random assignment to either *ex ante* or *ex post* framing. As shown in Table 4, Column 7, subjects who were asked to set a fine for a specific individual already found guilty of littering imposed penalties that were, on average, \$150 lower. This represents a 14% reduction compared to fines set by subjects asked to determine a general penalty before any violation had occurred.¹⁸ This result suggests that while penalties are often designed with deterrence in mind, decision-makers may *ex post* adjust penalties downward when considering the real-world consequences for offenders. This misalignment between *ex ante* and *ex post* judgments may help explain why penalties can be set too high: they are calibrated to deter but ultimately exceed what people believe is fair or appropriate for individual violators. Such a disconnect underscores their potential to backfire by discouraging the reporting of wrongdoing.

4.2 Implications on Crime

¹⁷Appendix Table A2 confirms these results are robust to alternative specifications of the underlying preference measures.

¹⁸Table 4, Column 8, confirms that the results are robust to outliers by limiting the sample to subjects who suggested a fine between 500 and 4000 USD, the recommended range provided in the prompt.

Table 5: Effect of Penalty Severity and Reporting Probability on Managerial Misconduct

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Brk Law	Brk Law	Cond. Brk Law	Uncond. Brk Law	Prior Report	Prior Report	Belief Report
Prop. Rprtnng	-0.0686*** (0.00387)	-0.0676*** (0.00617)					
Fine		-0.149** (0.0745)	-0.183*** (0.0559)	-0.341*** (0.0657)			-0.000145 (0.0411)
Fine * Prop. Rprtnng		-0.00206 (0.00955)					
Prior Fine					0.114*** (0.0303)		
Prior 40% fine						0.0788*** (0.0158)	
Prior 100% fine						0.115*** (0.0292)	
Observations	2540	2540	254	255	765	765	255
Mean Brk Law (0 to 10% reported)	0.709	0.709					
Mean Brk Law (Fine 5%)			0.264	0.462			
Mean Prior Report (Fine 5%)					0.494	0.494	
Mean Belief Report (Fine 5%)							0.370

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Brk Law is an indicator for whether the manager chose to break the law for a given reporting range, Prop. Rprtnng. Cond. Brk Law is whether the manager chose to break the law for the reporting rate that was implemented in their session. Uncond. Brk Law is an indicator for whether the manager chose to break the law when they did not select based on expected reporting rates. Prior Report is the manager's belief about the proportion of participants in prior sessions who reported a law-breaking manager that was elicited in Stage 3 for given fine rates. Belief Report is the manager's belief about the proportion of participants in prior sessions who reported a law-breaking manager with their given fine rate elicited during Stage 2. Mean Brk Law (0–10% reported) is the average decision to break the law among managers in the 1–10% reporting probability scenario. Mean Brk (Fine 5%) is the mean dependent variable rate for managers assigned to the 5% fine rate. Mean Prior Report (Fine 5%) is the average belief managers held about the reporting rate under the 5% penalty condition elicited in Stage 3. Mean Belief Report (5%) is the average belief managers assigned to the 5% fine rate held about the reporting rate during Stage 2. Prop. Rprtnng is an ordinal variable ranging from 1 to 10, where each unit represents a 10 percentage point band of reporting rates observed in previous sessions: 1 corresponds to 0–10%, 2 to 11–20%, and so on, up to 10 corresponding to 91–100%. Fine and Prior Fine indicate the randomly assigned penalty level —5, 40, or 100 percent of the manager's earnings. Regressions include an indicator for whether priors were elicited simultaneously. Column 1 and 2 observations are equal to the number of managers multiplied by the ten reporting scenarios. Column 3, 4, and 7 observations are the number of managers. Column 5 and 6 observations are equal to the number of managers multiplied by the three fine scenarios. Standard errors are clustered at the manager level in Columns 1, 2, 5 and 6, and are robust otherwise.

In Table 5, Column 1, we find that managers are significantly less likely to break the law when the likelihood of being reported by employees is higher. For a 10pp increase in the likelihood of being reported by an employee, managers are 6.86pp less likely to have broken the law.¹⁹ Notably, this effect is not influenced by the size of the penalty — managers are no less likely to break the law for a given reporting probability, regardless of the penalty amount (Table 5, Column 2). However, when we hold the likelihood of reporting constant, we observe that higher penalties do reduce law-breaking: for every 10pp increase in the fine, managers are 1.49pp less likely to decide to break the law.

Our earlier finding showed that the likelihood of reporting decreases as penalties increase. We now also find that managers are *less likely* to break the law when penalties are more severe, but *more likely* to break the law when the probability of being reported is lower. As a result, the overall impact of penalty severity on law-breaking behavior depends on the balance between these opposing forces, even when managers have information about the probability of being reported.

In our experiment, we observe that despite the reduction in reporting, higher penalties

¹⁹Specifically, managers are asked about their likelihood of breaking the law for a 10pp range in the probability of being reported from the previous day's session.

ultimately decrease law-breaking. To illustrate, consider the effect of increasing a fine from 0 to 100 percent. This change reduces law-breaking by 14.9 pp (Table 5, Column 2) but also reduces reporting by 9.8pp (Table 2, Column 1). This decrease in reporting, in turn, leads to an approximate 6.7pp increase in law-breaking (Table 5, Column 2). Consequently, the net effect is a reduction in law-breaking of 8.2pp.²⁰ We confirm this by estimating only those decisions that we implemented (which were a function of the proportion of people who had reported in previous sessions), and we do indeed find that at higher penalties, managers were still less likely to break the law (Table 5, Column 3). Importantly, both managers and employees anticipated this relationship: subjects consistently expected lower levels of law-breaking at higher penalties, as shown in Appendix Table A3, further reinforcing the credibility of the observed behavioral response.

When managers lack information about the probability of being reported, they are even more likely to reduce their law-breaking behavior in response to higher penalties (Table 5, Column 4). This effect may be consistent with their priors regarding how penalties influence reporting. Using our within-subject, incentive-compatible elicitation of managers' priors, we find that managers expected higher penalties to increase the likelihood of employees reporting them. Specifically, they predicted that a 10pp increase in the penalty rate would lead to a 1.14pp increase in reporting (Table 5, Column 5 and 6). Based on these beliefs, both the penalties the managers would face and their perceived probability of being caught align in a way that magnifies the deterrent effect: they are motivated to reduce law-breaking because they both fear harsher penalties and believe that the chances of getting caught increase with higher penalties. However, this result does not hold when considering across-subject variation, where we find no significant relationship between the randomly assigned fines managers face and their expectations on the probability of being reported (Table 5, Column 7).

While we cannot definitively conclude that managers believe higher fines directly lead to increased reporting, our results consistently suggest that managers did not expect more severe penalties to *reduce* the likelihood of being reported. These findings imply that, in the absence of information about the probability of being caught, potential law-breakers may initially worry that harsher penalties could increase the likelihood of reporting and enforcement. However, once they gain more insight into actual reporting rates—regardless of their beliefs about what influences those rates—they are likely to adjust their behavior accordingly. Specifically, as reporting rates decrease, they become more inclined to engage in misconduct. Indeed, as we've observed, once managers learn about reporting rates, they do adjust their behavior: when reporting is lower, they are more likely to break the law, and when penalties increase, reporting decreases.

²⁰We use the example of a fine increasing from 0 to 100 percent to allow for direct interpretation of the coefficients from the table. However, the same general pattern holds when comparing the 5% penalty to the 100% penalty.

We may wonder why managers' priors contrast with the priors and behaviors of employees, who anticipated and exhibited reduced reporting at higher penalties. One explanation is that the discrepancy arises because managers were primed to think about the consequences of their actions, while employees had direct experience with the decision to report. Overall, these results highlight a nuanced relationship between penalty rates and beliefs about reporting, with the effects differing depending on whether the subject is deciding to break the law or report the wrongdoing.

5 Conclusion

This study provides new causal evidence that harsher penalties, while intended to deter misconduct, can inadvertently suppress a critical component of enforcement: the willingness to report wrongdoing. Across both incentivized experimental decisions and hypothetical scenarios, we find that individuals are consistently less likely to report violations when the consequences for the perpetrator are more severe. This reduction in reporting is strongly mediated by perceptions of fairness: individuals are less willing to enforce penalties they view as excessive. This finding may help reconcile earlier lab work in which penalty severity reduced jury convictions in some cases but not others, possibly because some penalties were perceived as fair while others were not (e.g., (Kerr, 1978)).

Importantly, this suppression of reporting has meaningful implications for deterrence. In our setting, reduced reporting partially offsets, but does not fully eliminate, the deterrent effect of higher penalties. On net, harsher punishments still reduce misconduct, but only because their direct deterrent effect outweighs the indirect increase in violations caused by lower reporting rates. In settings where detection relies more heavily on reporting, or where penalties are even more misaligned with perceived fairness, this balance could reverse. Indeed, in some hypothetical scenarios, reporting falls by more than 50pp under the most severe penalties, suggesting that real-world enforcement could be weakened in environments with especially punitive consequences.

Our experimental design intentionally isolates the direct effect of punishment severity on reporting by turning off several real-world complexities: we remove ambiguity about guilt which rules out the possibility of false convictions, and eliminate strategic beliefs about how others in the enforcement chain might behave. While removing these features improves our understanding of key mechanisms, one might worry that our findings will not hold when such complications are present. However, both false convictions and beliefs about how others will act would likely amplify the reluctance to report under high penalties. The fear of punishing an innocent person, the belief that others will abstain from enforcement, or the hope that someone else will step in, are all factors that would likely further suppress reporting. As a result, the suppression effects we identify in this simplified setting are likely to be lower bounds. In real-world environments,

where fairness concerns are compounded by uncertainty and social interdependence, the unintended effect of severe penalties on reporting may be even stronger.

Our findings also help explain why overly harsh penalties persist, despite reluctance to enforce them in practice. We show that individuals support more severe punishments when acting *ex ante*, when deterrence is relevant, but prefer more lenient consequences once the violation has occurred and punishment must be imposed. This disconnect contributes to an institutional design problem: penalties may be calibrated to deter but then go unenforced, weakening both deterrence and legitimacy. By documenting this gap experimentally, we provide a behavioral foundation for understanding how enforcement systems become overly punitive on paper while remaining selectively or inconsistently applied in practice.

More broadly, our findings contribute to a growing literature on how social norms and perceptions of fairness shape the effectiveness of legal and institutional systems. When rules or penalties conflict with prevailing fairness norms, they may be undermined by those responsible for enforcing them. Ultimately, the effectiveness of a legal or institutional sanction depends not only on the consequences of wrongdoing, but also on whether people are willing to uphold those consequences. Our results underscore the importance of calibrating sanctions not only to the severity of the offense, but also to the social acceptability of enforcement. By aligning penalties with the public's willingness to report, policymakers can avoid inadvertently weakening enforcement and can design systems that are both effective and socially sustainable.

References

- Acemoglu, D., & Jackson, M. O. (2017). Social norms and the enforcement of laws. *Journal of the European Economic Association*, 15(2), 245–295.
- Aimone, J., Rentschler, L., Smith, V., & Wilson, B. J. (2025). Sympathy with resentment: Willingness to report criminal behavior depends on the punishment. *Public Choice*, 202, 343–365. <https://doi.org/10.1007/s11127-024-01233-6>
- Alekseev, A., Charness, G., & Gneezy, U. (2017). Experimental methods: When and why contextual instructions are important. *Journal of Economic Behavior & Organization*, 134, 48–59. <https://doi.org/https://doi.org/10.1016/j.jebo.2016.12.005>
- Becker, G. S. (1968). Crime and punishment: An economic approach. *Journal of Political Economy*, 76(2), 169–217.
- Bindler, A., & Hjalmarsson, R. (2018). How punishment severity affects jury verdicts: Evidence from two natural experiments. *American Economic Journal: Economic Policy*, 10(4), 36–78.
- Bjerk, D. (2005). Making the crime fit the penalty: The role of prosecutorial discretion under mandatory minimum sentencing. *The Journal of Law and Economics*, 48(2), 591–625.
- Brandts, J., & Charness, G. (2000). Hot vs. cold: Sequential responses and preference stability in experimental games. *Experimental Economics*, 2(3), 227–238.
- Brandts, J., & Charness, G. (2011). The strategy versus the direct-response method: A first survey of experimental comparisons. *Experimental Economics*, 14(3), 375–398.
- Brank, E. M., Woolard, J. L., Brown, V. E., Fondacaro, M., Luescher, J. L., Chinn, R. G., & Miller, S. A. (2007). Will they tell? weapons reporting by middle-school youth. *Youth Violence and Juvenile Justice*, 5(2), 125–146.
- Brinkley, C. J., & Saarnio, D. A. (2006). Involving students in school violence prevention: Are they willing to help? *Journal of School Violence*, 5(1), 93–106.
- Burdea, V., & Woon, J. (2022). Online belief elicitation methods. *Journal of Economic Psychology*, 90, 102496.
- Butler, J. V., Serra, D., & Spagnolo, G. (2020). Motivating whistleblowers. *Management Science*, 66(2), 605–621.
- Chin, Y.-M., & Cunningham, S. (2019). Revisiting the effect of warrantless domestic violence arrest laws on intimate partner homicides. *Journal of Public Economics*, 179, 104072. <https://doi.org/10.1016/j.jpubeco.2019.104072>
- Danz, D., Vesterlund, L., & Wilson, A. J. (2020). *Belief elicitation: Limiting truth telling with information on incentives* (tech. rep.). National Bureau of Economic Research.

- Devine, D. J. (2012). *Jury decision making: The state of the science*. New York University Press.
- Drysdale, D. A. (2010). *Campus attacks: Targeted violence affecting institutions of higher education*. DIANE Publishing.
- Dyck, A., Morse, A., & Zingales, L. (2010). Who blows the whistle on corporate fraud? *The journal of finance*, 65(6), 2213–2253.
- Falk, A., Becker, A., Dohmen, T., Huffman, D., & Sunde, U. (2023). The preference survey module: A validated instrument for measuring risk, time, and social preferences. *Management Science*, 69(4), 1935–1950.
- Hamby, S., Weber, M. C., Grych, J., & Banyard, V. (2016). What difference do bystanders make? the association of bystander involvement with victim outcomes in a community sample. *Psychology of violence*, 6(1), 91.
- Hart, T. C., & Miethe, T. D. (2008). Exploring bystander presence and intervention in nonfatal violent victimization: When does helping really help? *Violence & Victims*, 23(5).
- Iyengar, R. (2009). Does the certainty of arrest reduce domestic violence? evidence from mandatory and recommended arrest laws. *Journal of public Economics*, 93(1-2), 85–98.
- Kahan, D. M. (2000). Gentle nudges vs. hard shoves: Solving the sticky norms problem. *The University of Chicago Law Review*, 607–645.
- Kerr, N. L. (1978). Severity of prescribed penalty and mock jurors' verdicts. *Journal of Personality and Social Psychology*, 36(12), 1431–1442. <https://doi.org/10.1037/0022-3514.36.12.1431>
- Lacasse, C., & Payne, A. A. (1999). Federal sentencing guidelines and mandatory minimum sentences: Do defendants bargain in the shadow of the judge? *The Journal of Law & Economics*, 42(S1), 245–270. <https://doi.org/10.1086/467425>
- Philpot, R., Liebst, L. S., Levine, M., Bernasco, W., & Lindegaard, M. R. (2020). Would i be helped? cross-national cctv footage shows that intervention is the norm in public conflicts. *American Psychologist*, 75(1), 66.
- Research, N. S. (2023). British social attitudes survey 2023: Full questionnaire [Accessed: 2025-03-20].
- Smith, A. (1978). *Lectures on jurisprudence* (R. L. Meek, D. D. Raphael, & P. G. Stein, Eds.). Liberty Fund.
- Vidmar, N. (1972). Effects of decision alternatives on the verdicts and social perceptions of simulated jurors. *Journal of Personality and Social Psychology*, 22(2), 211–218.

A Appendix Tables

Table A1: Non-Completion by Treatment and Firm Role

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Employee	Manager	5% fine	40% fine	100% fine	Mngr. = Empl.	5% = 40% + 100%	40% = 5% + 100%	100% = 5% + 40%
Mean Non-Completion	0.05	0.15	0.06	0.08	0.08	-0.10***	0.02	-0.01	-0.01
Observations	1379	327	576	544	598	1706	1718	1718	1718

Columns 1 to 5 report the mean non-completion rate by treatment arm and firm role. Columns 6 to 9 report t-statistics from tests of equality between groups: Column 6 compares employees and managers; Columns 7 to 9 compare each treatment arm against the other two.

Table A2: Punishment Severity and Reporting Rates, Detailed

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Reported	Reported	Reported	Reported	Reported	Reported	Prior Report	Prior Report	Reported
Fine	-0.0981*** (0.0329)		-0.0891*** (0.0332)	-0.0856** (0.0335)	-0.0872*** (0.0336)	-0.0886*** (0.0323)			-0.0844*** (0.0322)
40% fine		-0.0106 (0.0309)							
100% fine		-0.0893*** (0.0314)							
Female				0.0404 (0.0270)	0.0410 (0.0271)	0.0211 (0.0260)			0.0288 (0.0263)
Age				0.00101 (0.00112)	0.000749 (0.00114)	0.000130 (0.00109)			0.0000399 (0.00110)
Non-white				-0.0311 (0.0270)	-0.0362 (0.0272)	-0.000173 (0.0263)			-0.0327 (0.0262)
Altruism Index					0.00894 (0.0182)	0.0123 (0.0177)			
Neg. Rec. Index					0.000897 (0.0160)	-0.00861 (0.0153)			
Pos. Rec. Index					-0.00229 (0.0184)	-0.0215 (0.0185)			
Law-Abiding Reluctance					-0.0229* (0.0120)	-0.0286** (0.0116)			
Apprt. Penalty						0.00397*** (0.000374)			
Prior Fine							-0.0805*** (0.0145)		
Simultaneous Priors							-0.0212* (0.0113)	-0.0212* (0.0113)	
Prior 40% fine								-0.0232*** (0.00844)	
Prior 100% fine								-0.0757*** (0.0141)	
Avg. Report Cheating									0.146*** (0.0437)
Avg. Report Illegal Parking									0.0952** (0.0394)
Report Discrimination									0.176*** (0.0395)
Report Vandalism									0.0599** (0.0289)
Day Fixed Effects	No	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1147	1147	1147	1132	1129	1128	3429	3429	1129
Mean Reported (Fine 5%)	0.777	0.777	0.777	0.777	0.777	0.777			0.777
Mean Prior Report (Fine 5%)							0.600	0.600	

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Reported is a binary variable equal to 1 if the employee chose to report their manager in the Main Experiment. Prior Report is the employee's belief about the proportion of participants in prior sessions who reported a law-breaking manager. Mean Reported (Fine 5%) is the mean reporting rate for employees assigned to the 5% treatment arm. Mean Prior Report (Fine 5%) is the average belief employees held about the reporting rate under the 5% penalty condition. Fine and Prior Fine indicate the randomly assigned penalty level—5, 40, or 100 percent of the manager's earnings. (Prior) 40% Fine and 100% Fine are indicators for the respective fine rate. Demographics are Age, Female, and Non-White. Altruism Index, Positive Reciprocity Index, Negative Reciprocity Index, and Law Abiding Reluctance refer to elicited subject preferences. Apprt. Penalty refers to the penalty level the employee reported as appropriate for a manager who breaks the law. Simultaneous Priors is an indicator if the subject was randomly assigned to report priors on law-breaking and reporting simultaneously. Avg. Cheating, Avg. Report Illegal Parking, Report Discrimination, and Report Vandalism are each subject-level averages indicating whether the respondent would report the respective violation, averaged across all penalty levels they were shown. Columns 1–6 and 9 include one observation per employee. Columns 7–8 include three observations per employee, one for each fine level. Columns 7–8 include an indicator for whether priors were elicited simultaneously. Standard errors are clustered at the individual level in Columns 7 and 8, and robust otherwise.

Table A3: Hypothetical Reporting Scenarios, Detailed

	(1) Reported	(2) Reported	(3) Reported	(4) Reported	(5) Reported	(6) Reported
Discrimination: warned and monitor	-0.0206 (0.0260)					
Discrimination: suspended	-0.115*** (0.0287)					
Discrimination: fired	-0.0574** (0.0273)					
Cheating: reduced grade		-0.0414** (0.0169)				
Cheating: zero		-0.240*** (0.0177)				
Cheating: fail class		-0.411*** (0.0172)				
Cheating: fail and suspended		-0.513*** (0.0162)				
Cheating: expell from school		-0.554*** (0.0157)				
Vandalism: pay for the damages (200), 100 hrs service			-0.0731** (0.0311)	0.0651** (0.0316)		
Vandalism: pay for the damages (200), 100 hrs service and 2 mnths jail			-0.275*** (0.0312)	0.0850** (0.0382)		
Vandalism: pay for the damages (200), 100 hrs service * 1[Excessive]				-0.520*** (0.0449)		
Vandalism: pay for the damages (200), 100 hrs service and 2 mnths jail * 1[Excessive]				-0.572*** (0.0394)		
Parking: 500 fine					-0.234*** (0.0119)	0.0603*** (0.0182)
Parking: 1000 fine					-0.435*** (0.0138)	0.0518** (0.0248)
Parking: 2000 fine					-0.505*** (0.0139)	0.0232 (0.0317)
Penalty Illegal Parking: 500 fine * 1[Excessive]						-0.618*** (0.0211)
Penalty Illegal Parking: 1,000 fine * 1[Excessive]						-0.627*** (0.0250)
Penalty Illegal Parking: 2,000 fine * 1[Excessive]						-0.613*** (0.0315)
Observations	1399	8412	1401	1401	5611	5610
Mean Reported (Least Severe)	0.874	0.749	0.708	0.708	0.764	0.764

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Reported is a binary variable equal to 1 if the subject chose to report the violator for the corresponding penalty in the scenario. Mean Reported (Least Severe) is the average reporting rate under the least severe penalty. Dependent variables are indicators for the described penalty and scenario. 1[Excessive] is an indicator equal to 1 if the subject judged the given penalty to be excessive. Observations equal the number of subjects multiplied by the number of penalties shown per scenario. Standard errors are clustered at the individual level.

Table A4: Punishment Severity and Reporting Rates Heterogeneity, Detailed

	(1) Reported	(2) Reported	(3) Reported	(4) Reported	(5) Reported	(6) Reported	(7) Reported
Fine	-0.179 (0.117)	-0.0901** (0.0423)	-0.0555 (0.0701)	-0.00717 (0.0801)	-0.0899* (0.0508)	-0.453** (0.217)	-0.110 (0.0927)
Altruism, Self-Reported	-0.00137 (0.00847)						
Fine * Altruism, Self-Reported	0.0103 (0.0143)						
Altruism, Money Donation		0.000110 (0.000108)					
Fine * Altruism, Money Donation		-0.0000663 (0.000195)					
Neg. Rec., Self			0.00244 (0.00766)				
Fine * Neg. Rec., Self			-0.00845 (0.0125)				
Neg. Rec., Other				0.0123 (0.00774)			
Fine * Neg. Rec., Other				-0.0160 (0.0128)			
Neg. Rec., Revenge					-0.00948 (0.00818)		
Fine * Neg. Rec., Revenge					-0.00236 (0.0135)		
Pos. Rec., Self-Reported						-0.0205 (0.0148)	
Fine * Pos. Rec., Self-Reported						0.0399* (0.0240)	
Pos. Rec., Gift							0.00174 (0.0114)
Fine * Pos. Rec., Gift							0.00256 (0.0187)
Observations	1145	1145	1146	1146	1145	1145	1146
Mean Reported (Fine 5%)	0.777	0.777	0.777	0.777	0.777	0.777	0.777

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Reported is a binary variable equal to 1 if the employee chose to report their manager in the Main Experiment. Mean Reported (Fine 5%) is the mean reporting rate for employees assigned to the 5% treatment arm. Fine indicate the randomly assigned penalty level — 5, 40, or 100 percent of the manager's earnings. Independent variables describe individual questions used to elicit preferences, corresponding to Altruism, Positive Reciprocity, and Negative Reciprocity. Standard errors are robust.

Table A5: Beliefs on Law-Breaking Under Different Penalty Rates

	(1)	(2)	(3)
	Prior BrkLaw	Prior BrkLaw	Prior BrkLaw
Prior 40% fine	-0.291*** (0.0174)	-0.269*** (0.00810)	-0.273*** (0.00734)
Prior 100% fine	-0.495*** (0.0271)	-0.507*** (0.0122)	-0.505*** (0.0111)
Sample	Manager	Employee	All
Mean Prior Report (Fine 5%)	0.699	0.694	0.695

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Prior BrkLaw is the elicited belief about the proportion of managers who broke the law in prior sessions under a given fine rate. Mean Prior BrkLaw(Fine 5%) is the average belief employees held about the rate at which managers would break the law under the 5% fine rate. Prior 40% Fine and 100% Fine are indicators for the respective fine rate. All regressions include an indicator for whether priors were elicited simultaneously. Standard errors are robust.